

Actuarial Corporate Finance

Lesson 4

Capital Asset Pricing Model

Prof. Prabhdeep Singh

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- ▶ By 9/4 end of day, send me e-mail with your topic and partner. Send me three possible time slots. Time slots are not guaranteed.

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Portfolio theory: In presence of a risk free asset, efficient portfolios are the highest sloped straight line from risk free asset to risky efficient frontier

Review of Homework

Measuring Systematic Risk of a Stock - Algebraic Derivation

$$\blacktriangleright \text{Var}(R_{\text{eff}}) = \sum_i \cdot x_i \cdot \text{Cov}(R_i, R_{\text{eff}}) = \sum_i \cdot x_i \cdot SD(R_i) \cdot SD(R_{\text{eff}}) \cdot \text{Corr}(R_i, R_{\text{eff}})$$

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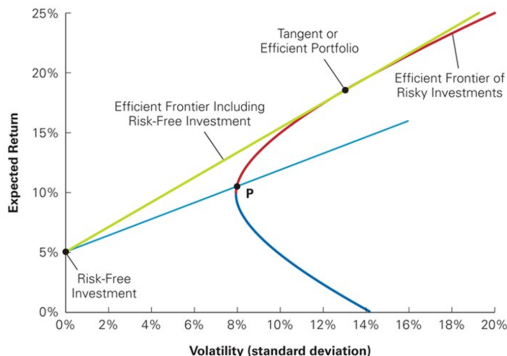
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- ▶ Almost!: $\mathbb{E}[R_i] - R_f = \beta_i \cdot (\mathbb{E}[R_{\text{eff}}] - R_f)$. Needs portfolio theory to get to this conclusion.

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Required Returns - the missing piece of the puzzle

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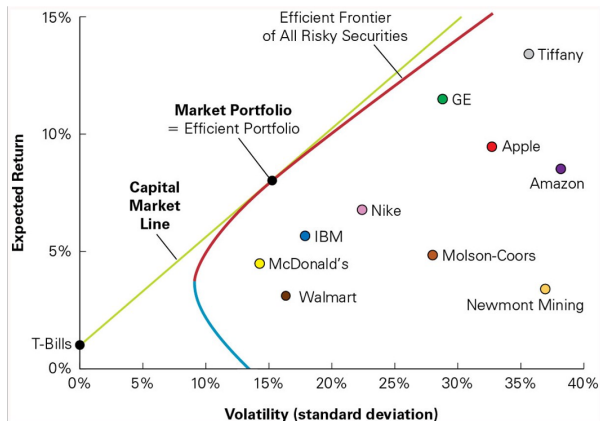
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- **Capital Market Line (CML)**: efficient frontier for market PF + risk-free asset
- Optimal investment must be combo of market PF + risk-free asset (on CML)

Capital Market Line



► Slope of CML = Sharpe ratio of market PF

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- The **Beta** of stock i is defined w.r.t. the market PF:

$$\beta_i = \frac{\text{Cov}(R_i, R_M)}{\text{Var}(R_M)} =$$

- R_M = return of market PF
- For later: $\sigma_M = \rho_{i,M} \cdot \sigma_i / \beta_i$.

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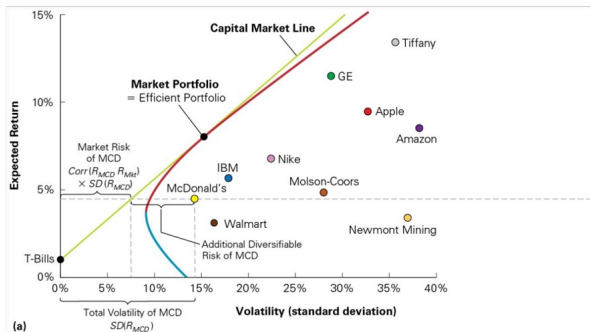
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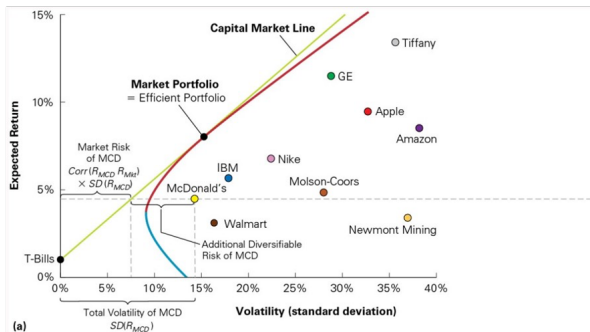
CML Equation

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 - Risk premium of i is proportional to its correlation with market PF
 - For given (expected) return, asset on CML has min. vola (max. Sharpe ratio)



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► Exercise W 4-2